

Blind DC Estimation in the Presence of Non-Gaussian Signal

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Abstract—A method of estimating DC offset based solely on the received signal, without using any training information, is presented. This method takes into account the non-Gaussian statistical distribution of the modulated desired signal in order to produce a DC estimate that is more accurate than the conventional sample average of the received signal. Specifically, two iterative algorithms are derived based on the generalized expectation-maximization algorithms for jointly estimating the DC offset, the noise variance, and the parameters that characterize the signal distribution. As an application, we consider the special case where the modulated signal has a constant-envelope characteristic. Using the GSM/EDGE air interface, we demonstrate that the proposed method yields significant reduction in the DC-offset estimation error, which translates into performance gains in receiver sensitivity when used in direct-conversion receivers.

Index Terms—DC Estimation, Non-Gaussian, Expectation Maximization, GSM, EDGE.

I. INTRODUCTION

The direct-conversion (or homodyne) receiver architecture has become increasingly popular for mobile terminals because of its power efficiency and cost effectiveness. Receivers with such an architecture convert the radio signal directly to baseband without using any intermediate frequencies and thus reduce the number of circuit components needed compared to the conventional heterodyne receivers. However, homodyne receivers often suffer from a DC-offset problem which is a by-product of the direct-conversion process. This DC offset often varies significantly from one transmission time interval or time slot to another such that it cannot be accurately estimated by a long-term average of the received signal over multiple time slots. Effective methods of estimating the DC offset independently over each time slot are therefore highly desirable.

One method proposed in [1] is to estimate the DC offset, jointly with the channel response, over a sequence of known training or pilot symbols transmitted in each time slot. Unfortunately, the accuracy of such training-based DC estimate is often limited by the small number of known symbols available in each time slot. Moreover, training sequences are not typically designed for estimating DC offset, and, as a result, the quality of the estimate may vary from one sequence to another, as is the case in the EDGE system [2].

In this paper, we consider methods of blindly estimating DC offset without using any training information. The simplest and most commonly used method of estimating DC offset blindly is to compute the average of the received signal over

each time slot. This estimate is optimal (in the maximum-likelihood sense) when both the desired signal and the noise are Gaussian-distributed. However, the distributions of digitally modulated signals are often far from Gaussian due to practical constraints on the constellation size and the peak-to-average power ratio (PAR) of the transmitted signal. For instance, the GMSK-modulated signal used in GSM/EDGE systems has constant amplitude, and hence its distribution has a compact support on a circle. In this paper, we show that the statistical distribution of the desired signal can be exploited to improve DC estimation.

II. SYSTEM MODEL

Let $\{r[n]\}_{n=1}^N$ denote a sequence of discrete-time received samples of length N . Our baseband model of $r[n]$ is given by

$$r[n] = s_\alpha[n] + d + w[n], \quad (1)$$

for $n = 1, 2, \dots, N$, where $s_\alpha[n]$ denotes the desired transmitted signal, d denotes the (complex-valued) DC offset, and $\{w[n]\}$ denotes a white Gaussian process with variance σ^2 . We model $s_\alpha[n]$ as a deterministic function $f(\cdot, \alpha)$ of an underlying random innovation process $\{u[n]\}$ with a real-valued parameter (or a vector of parameters) α which lies in a certain admissible set Ω , i.e. $s_\alpha[n] \equiv f(u[n], \alpha)$. We assume that the innovation process $\{u[n]\}$ is stationary with a certain known, absolutely continuous probability distribution that is independent of the parameter α and the noise process $\{w[n]\}$.

To simplify notations, let $\mathbf{r} = (r[1], r[2], \dots, r[N])^T$ and $\mathbf{u} = (u[1], u[2], \dots, u[N])^T$. Also let

$$\overline{x[n]} \equiv \frac{1}{N} \sum_{n=1}^N x[n] \quad \text{and} \quad \text{SV}\{x[n]\} \equiv \overline{|x[n] - \overline{x[n]}|^2}$$

denote the sample average and the (biased) sample variance of $x[n]$, respectively, for any sequence $x[n]$ of length N .

III. MAXIMUM-LIKELIHOOD BLIND DC ESTIMATION

A. Gaussian Signaling

If the transmitted signal $s_\alpha[n]$ is Gaussian distributed, the likelihood function of the DC offset d given the received samples $\{r[n]\}_{n=1}^N$ is simply the Gaussian probability density function given by

$$p(\mathbf{r}|d, \alpha, \sigma, \sigma_s^2) = \frac{1}{(\pi(\sigma_s^2 + \sigma^2))^N} e^{-\frac{\sum_{n=1}^N |r[n] - s_\alpha[n] - d|^2}{\sigma_s^2 + \sigma^2}}$$

where $\sigma_s^2 \equiv E|s_\alpha[n]|^2$ denotes the variance of the desired signal. It follows that the maximum-likelihood (ML) estimate of d is simply the received-signal average given by

$$\hat{d}_{ML} = \arg \min_d \sum_{n=1}^N |r[n] - d|^2 = \overline{r[n]}, \quad (2)$$

regardless of what the signal variance σ_s^2 and the noise variance σ^2 are. However, in practice, the distribution of the transmitted signal is often far from Gaussian. In this case, a better blind DC estimate can be derived by exploiting the non-Gaussian distribution of the signal, as shown below.

B. Non-Gaussian Signaling

From model (1), the likelihood function of the DC offset d is given by

$$p(\mathbf{r}|d, \alpha, \sigma) = \frac{1}{(\pi\sigma^2)^N} \int e^{-\sum_{n=1}^N \frac{|\mathbf{r} - f(u[n], \alpha) - d|^2}{\sigma^2}} p(\mathbf{u}) d\mathbf{u} \quad (3)$$

where $p(\mathbf{u})$ denotes the joint probability density function of the innovation vector $\mathbf{u}$. Setting the partial derivative of (3) with respect to d to zero, we obtain a necessary condition for the ML estimate of DC offset to satisfy for any given α and σ ,

$$\hat{d}_{ML} = \overline{r[n]} - \overline{E[s_\alpha[n]|\mathbf{r}, \hat{d}_{ML}, \alpha, \sigma]}, \quad (4)$$

which can be viewed as a generalization of (2) to possibly non-Gaussian signals.

Equation (4) shows that the ML estimate of d in the presence of a non-Gaussian distributed signal can be obtained by subtracting an estimate of the contribution of the non-Gaussian signal, namely the averaged conditional mean of the transmitted signal given the received signal and various parameters, from the sample average of received signal. When the signal-to-noise ratio (SNR) is low (e.g. $\sigma_s^2/\sigma^2 \ll 1$), the contribution of the second term in (4) becomes negligible, in which case, the DC estimate reduces back to the sample average of the received signal.

Note that if both α and σ are known *a priori*, (4) can be used to compute $\hat{d}_{ML}$ iteratively as

$$\hat{d}^{(m+1)} = \overline{r[n]} - \overline{E[s_\alpha[n]|\mathbf{r}, \hat{d}^{(m)}, \alpha, \sigma]},$$

where $\hat{d}^{(m)}$ denotes the DC estimate at the m -th iteration. In practice, however, α and σ also need to be estimated along with d .

IV. ITERATIVE JOINT ESTIMATION ALGORITHMS

In this section, we consider the ML joint estimate of the DC offset d , the noise variance σ^2 and the parameter α given $\mathbf{r}$. More precisely, we want to compute

$$(\hat{d}, \hat{\alpha}, \hat{\sigma}) = \arg \max_{(d, \alpha, \sigma)} \log p(\mathbf{r}|d, \alpha, \sigma). \quad (5)$$

One can solve this maximization problem using traditional derivative-based methods by taking partial derivatives with respect to the parameters α and σ , which would yield, along with (4), a system of three nonlinear equations with three unknowns, namely (d, α, σ) . Unfortunately, the solutions of these equations do not have closed-form expressions in general.

Here we present two effective algorithms for jointly estimating (d, α, σ) . Both algorithms are iterative in nature and are derived based on the classical theory of Expectation-Maximization (EM). The first algorithm corresponds to the original EM algorithm, while the second algorithm corresponds to a generalized EM algorithm. Both algorithms converge in the sense that the likelihood of the resulting joint estimate monotonically increases after each iteration.

Let $(\hat{d}^{(m)}, \hat{\alpha}^{(m)}, \hat{\sigma}^{(m)})$ denote the joint estimate at the m -th iteration. Also let $E[\cdot|m]$ and $\text{var}[\cdot|m]$ denote, respectively, the conditional expectation and the conditional variance operators given the received signal $\mathbf{r}$ and the joint estimate $(\hat{d}^{(m)}, \hat{\alpha}^{(m)}, \hat{\sigma}^{(m)})$ at the m -th iteration. The main steps of the proposed algorithms are outlined below.

Algorithm 1 for General Non-Gaussian Signal

- 1) Initialize $\hat{d}^{(0)}$, $\hat{\alpha}^{(0)}$ and $\hat{\sigma}^{(0)}$, and set iteration index $m = 0$
- 2) Estimate parameter α of non-Gaussian signal as

$$\hat{\alpha}^{(m+1)} = \arg \min_{\alpha \in \Omega} \left\{ \text{SV}\{r[n] - E[s_\alpha[n]|m]\} + \overline{\text{var}[s_\alpha[n]|m]} \right\} \quad (6)$$

- 3) Estimate DC offset d as

$$\hat{d}^{(m+1)} = \overline{r[n]} - \overline{E[s_{\hat{\alpha}^{(m+1)}}[n]|m]}, \quad (7)$$

- 4) Estimate noise variance σ^2 as

$$(\hat{\sigma}^{(m+1)})^2 = \text{SV}\{r[n] - E[s_{\hat{\alpha}^{(m+1)}}[n]|m]\} + \overline{\text{var}[s_{\hat{\alpha}^{(m+1)}}[n]|m]} \quad (8)$$

- 5) Increment m and return to Step 2 unless certain stopping criteria, such as $m > M$ for some pre-defined maximum number of iterations M , is satisfied.

The second algorithm is similar to the first except that d is updated before α in each iteration.

Algorithm 2 for General Non-Gaussian Signal

- 1) Initialize $\hat{d}^{(0)}$, $\hat{\alpha}^{(0)}$ and $\hat{\sigma}^{(0)}$, and set iteration index $m = 0$
- 2) Estimate DC offset d as

$$\hat{d}^{(m+1)} = \overline{r[n]} - \overline{E[s_{\hat{\alpha}^{(m)}}[n]|m]}, \quad (9)$$

- 3) Estimate parameter α of non-Gaussian signal as

$$\hat{\alpha}^{(m+1)} = \arg \max_{\alpha \in \Omega} \left\{ \overline{2\text{Re}\left\{ E[s_\alpha[n]|m] \left(r[n] - \hat{d}^{(m+1)} \right)^* \right\}} - \overline{E[|s_\alpha[n]|^2|m]} \right\} \quad (10)$$

- 4) Estimate noise variance σ^2 as

$$(\hat{\sigma}^{(m+1)})^2 = \overline{\left| r[n] - \hat{d}^{(m+1)} - E[s_{\hat{\alpha}^{(m+1)}}[n]|m] \right|^2} + \overline{\text{var}[s_{\hat{\alpha}^{(m+1)}}[n]|m]} \quad (11)$$

- 5) Increment m and return to Step 2 unless certain stopping criteria, such as $m > M$ for some pre-defined maximum number of iterations M , is satisfied.

V. DERIVATIONS

In this section, we derive the EM-based joint estimation algorithms presented in Section IV. For completeness, we begin with a brief review of the classical EM theory.

A. Review of Expectation-Maximization Algorithm

The EM algorithm [3] addresses the problem of computing ML estimates of certain parameters of interest when a certain part of the observed data is “missing”. The ML estimates in such case often cannot be expressed in closed forms. The EM algorithm provides a generic, iterative method for computing the ML estimates from the incomplete observed data.

More precisely, let Y denote the observed (incomplete) data, X denote a missing part of the data, and θ denote the parameter to be estimated. The EM algorithm seeks to obtain the ML estimate of θ given Y by iteratively computing

$$\hat{\theta}^{(m+1)} = \arg \max_{\theta} Q(\theta | \hat{\theta}^{(m)}), \quad (12)$$

where $\theta^{(m)}$ denotes the estimate of θ at the m -th iteration,

$$\begin{aligned} Q(\theta | \hat{\theta}^{(m)}) &\equiv E \left[\log p(X, Y | \theta) | Y, \hat{\theta}^{(m)} \right] \\ &= \int p(X | Y, \hat{\theta}^{(m)}) \log p(X, Y | \theta) dX, \end{aligned} \quad (13)$$

$p(X, Y | \theta)$ denotes the likelihood function of θ given the complete data (X, Y) , and $p(X | Y, \theta)$ denotes the conditional probability density function of X given Y and θ .

It was shown in [3] that the recursion in (12) always converges in the sense that the sequence of resulting likelihood functions, $\{p(Y | \theta^{(m)})\}$, is monotonically non-decreasing, regardless of what the choice of X is. In fact, the monotonicity is preserved so long as the updated parameter $\hat{\theta}^{(m+1)}$ at each iteration is chosen such that

$$Q(\hat{\theta}^{(m+1)} | \hat{\theta}^{(m)}) \geq Q(\hat{\theta}^{(m)} | \hat{\theta}^{(m)}), \quad (14)$$

for all m . Clearly, (12) implies (14). The class of algorithms that yield a sequence of parameter estimates $\{\hat{\theta}^{(m)}\}$ that satisfies (14) is called the *generalized* EM algorithms [3]. To simplify notations, we let $Q^{(m)}(\theta) \equiv Q(\theta | \hat{\theta}^{(m)})$ in the sequel.

B. Formulation

For the problem in (5), we consider the received signal $\mathbf{r}$ as the observed incomplete data and the non-Gaussian innovation $\mathbf{u}$ as the missing data (i.e. $Y = \mathbf{r}$ and $X = \mathbf{u}$) in the formulation of the EM algorithm. Our parameter vector of interest is $\theta = (d, \alpha, \sigma)$. Thus, from (13), we have

$$\begin{aligned} Q^{(m)}(d, \alpha, \sigma) &= E \left[\log p(\mathbf{r}, \mathbf{u} | d, \alpha, \sigma) | \mathbf{r}, \hat{d}^{(m)}, \hat{\alpha}^{(m)}, \hat{\sigma}^{(m)} \right] \\ &= E [\log p(\mathbf{r} | \mathbf{u}, d, \alpha, \sigma) | m] + E [\log p(\mathbf{u}) | m]. \end{aligned} \quad (15)$$

Note that the second term in (15) is just a constant and can be neglected when optimizing over (d, α, σ) . Since $\{w[n]\}$ in (1) is assumed white and Gaussian, we have

$$p(\mathbf{r} | \mathbf{u}, d, \alpha, \sigma) = \frac{1}{(\pi\sigma^2)^N} \exp \left\{ -\frac{\sum_{n=1}^N |r[n] - s_{\alpha}[n] - d|^2}{\sigma^2} \right\}. \quad (16)$$

Substituting (16) into (15), we obtain

$$\begin{aligned} Q^{(m)}(d, \alpha, \sigma) &= -N \frac{E \left[|r[n] - s_{\alpha}[n] - d|^2 | m \right]}{\sigma^2} - N \log \pi \sigma^2 \\ &= -\frac{N}{\sigma^2} \left(|r[n] - E[s_{\alpha}[n] | m] - d|^2 + \text{var}[s_{\alpha}[n] | m] \right) \\ &\quad - N \log \pi \sigma^2. \end{aligned} \quad (17)$$

C. Parameter Optimization

We first derive Algorithm 1 presented in Section IV. The algorithm follows directly from the original EM recursion (12), which, based on the above formulation, can be written as

$$\left(\hat{d}^{(m+1)}, \hat{\alpha}^{(m+1)}, \hat{\sigma}^{(m+1)} \right) = \arg \max_{(d, \alpha, \sigma)} Q^{(m)}(d, \alpha, \sigma)$$

where the maximization is performed under the constraints of $\alpha \in \Omega$ and $\sigma \geq 0$, and where $Q^{(m)}(d, \alpha, \sigma)$ is given in (17). The best choice of (d, α, σ) can be obtained by sequentially optimizing $Q^{(m)}(d, \alpha, \sigma)$ over each of its three variables while keeping the remaining variables fixed, as described below.

Clearly, for any fixed α and σ , the best d that maximizes (17) is

$$\hat{d}(\alpha) \equiv \overline{r[n] - E[s_{\alpha}[n] | m]}, \quad (18)$$

which depends on the given α but not σ . It is also easy to show that for any fixed α and d , the best σ that maximizes (17) is given by

$$\hat{\sigma}(d, \alpha)^2 \equiv \overline{|r[n] - E[s_{\alpha}[n] | m] - d|^2} + \overline{\text{var}[s_{\alpha}[n] | m]}. \quad (19)$$

Substituting (18) into (19), we obtain the best σ for any fixed α as

$$\begin{aligned} \hat{\sigma}(\alpha)^2 &\equiv \hat{\sigma}(\hat{d}(\alpha), \alpha)^2 \\ &= \text{SV}\{r[n] - E[s_{\alpha}[n] | m]\} + \overline{\text{var}[s_{\alpha}[n] | m]}. \end{aligned} \quad (20)$$

By substituting (18) and (20) into (17), it follows that the best α is given by

$$\begin{aligned} \hat{\alpha}^{(m+1)} &\equiv \arg \max_{\alpha \in \Omega} Q^{(m)}(\hat{d}(\alpha), \alpha, \hat{\sigma}(\alpha)) \\ &= \arg \min_{\alpha \in \Omega} \log \hat{\sigma}(\alpha)^2 = \arg \min_{\alpha \in \Omega} \hat{\sigma}(\alpha)^2, \end{aligned}$$

which leads to (6). Now setting $\hat{d}^{(m+1)} = \hat{d}(\hat{\alpha}^{(m+1)})$ and $\hat{\sigma}^{(m+1)} = \hat{\sigma}(\hat{\alpha}^{(m+1)})$ yields (7) and (8), respectively.

The main difference between Algorithm 1 and 2 is that the DC estimate is updated using the estimate of α from the previous iteration instead of the current iteration. Specifically, Algorithm 2 is obtained by setting $\hat{d}^{(m+1)} = \hat{d}(\hat{\alpha}^{(m)})$,

$$\begin{aligned} \hat{\alpha}^{(m+1)} &\equiv \arg \max_{\alpha \in \Omega} Q^{(m)}(\hat{d}(\hat{\alpha}^{(m)}), \alpha, \hat{\sigma}(\hat{d}(\hat{\alpha}^{(m)}), \alpha)) \\ &= \arg \min_{\alpha \in \Omega} \hat{\sigma}(\hat{d}(\hat{\alpha}^{(m)}), \alpha)^2, \end{aligned} \quad (21)$$

and $\hat{\sigma}^{(m+1)} = \hat{\sigma}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m+1)})$, which leads to (9), (10) and (11), respectively. It follows from (21) and the definitions of $\hat{d}(\alpha)$ and $\hat{\sigma}(d, \alpha)$ in (18) and (19), respectively, that

$$\begin{aligned} Q^{(m)}(\hat{d}^{(m+1)}, \hat{\alpha}^{(m+1)}, \hat{\sigma}^{(m+1)}) &= Q^{(m)}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m+1)}, \hat{\sigma}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m+1)})) \\ &\geq Q^{(m)}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m)}, \hat{\sigma}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m)})) \\ &\geq Q^{(m)}(\hat{d}(\hat{\alpha}^{(m)}), \hat{\alpha}^{(m)}, \hat{\sigma}^{(m)}) \\ &\geq Q^{(m)}(\hat{d}^{(m)}, \hat{\alpha}^{(m)}, \hat{\sigma}^{(m)}) \end{aligned}$$

which satisfies (14). Therefore, Algorithm 2 also converges in the sense of monotonically increasing likelihoods.

VI. APPLICATION FOR CONSTANT-ENVELOPE SIGNALING

In this section, we consider the special case where the transmitted signal has constant but unknown amplitude. The key benefit of constant-envelope signaling is the resulting low peak-to-average power ratio of the transmitted signal that allows the power amplifier to operate at high efficiency. One well-known example of constant-envelope signaling schemes is Gaussian-Minimum Shift Keying (GMSK) [4], as adopted in the GSM/EDGE systems.

In this case, we model the transmitted signal as $s_\alpha[n] = f(u[n], \alpha) = \alpha \exp\{j u[n]\}$, where the innovation process $\{u[n]\}$ is a sequence of i.i.d. random variables that are uniformly distributed on $[-\pi, \pi)$ and where the parameter α is the signal amplitude with the admissible set $\Omega = [0, \infty)$. For this model, it was found that Algorithm 1 and Algorithm 2 yield virtually the same performance, while Algorithm 2 has lower implementation complexity. Due to space limitations, we will focus on Algorithm 2 in the sequel.

It is straightforward to show that $E[s_\alpha[n] | m] = \alpha \hat{v}^{(m)}[n]$ and $\text{var}[s_\alpha[n] | m] = \alpha^2(1 - |\hat{v}^{(m)}[n]|^2)$, where

$$\hat{v}^{(m)}[n] \equiv E[e^{j u[n]} | m] = e^{j \phi(r[n] - \hat{d}^{(m)})} \frac{I_1\left(\frac{2\hat{\alpha}^{(m)}|r[n] - \hat{d}^{(m)}|}{(\hat{\sigma}^{(m)})^2}\right)}{I_0\left(\frac{2\hat{\alpha}^{(m)}|r[n] - \hat{d}^{(m)}|}{(\hat{\sigma}^{(m)})^2}\right)}, \quad (22)$$

where $I_n(\cdot)$ is the n -th order modified Bessel function of the first kind, and $\phi(z)$ denotes the phase of z for any complex number z . Substituting these expressions back into (9), (10) and (11), along with some intuitive choices of initial estimates, leads to the following algorithm.

Algorithm for Constant-Envelope Signal

- 1) Initialize $\hat{d}^{(0)} = \overline{r[n]}$, $\hat{\alpha}^{(0)} = \overline{|r[n] - \hat{d}^{(0)}|}$, and $(\hat{\sigma}^{(0)})^2 = \overline{|r[n] - \hat{d}^{(0)}|^2} - (\hat{\alpha}^{(0)})^2$, and set iteration index $m = 0$
- 2) Compute estimate of complex phase exponentials $\{\hat{v}^{(m)}[n]\}_{n=1}^N$ according to (22),
- 3) Estimate DC offset d as

$$\hat{d}^{(m+1)} = \overline{r[n]} - \hat{\alpha}^{(m)} \overline{\hat{v}^{(m)}[n]},$$

- 4) Estimate signal amplitude α as

$$\hat{\alpha}^{(m+1)} = \text{Re} \left\{ \overline{\hat{v}^{(m)}[n] (r[n] - \hat{d}^{(m+1)})^*} \right\}^+$$

where $x^+ = x$ if $x \geq 0$ and $x^+ = 0$ if $x < 0$ for any real number x .

- 5) Estimate noise variance σ^2 as

$$(\hat{\sigma}^{(m+1)})^2 = \overline{|r[n] - \hat{d}^{(m+1)}|^2} - (\hat{\alpha}^{(m+1)})^2 \quad (23)$$

- 6) Increment m and return to Step 2 unless certain stopping criteria is satisfied.

Using the Cauchy-Schwarz inequality, it is easy to show that the right-hand side of (23) is always non-negative.

VII. NUMERICAL RESULTS

The GSM/EDGE air interface [6] was used to evaluate the performance of the proposed blind DC estimation algorithm, as described in Section VI. Specifically, the performance of the algorithm is investigated in the presence of a GMSK modulated signal. It is known that GMSK is a special variant of Frequency Shift Keying (FSK), cf. [5], which generates transmitted signals with constant amplitude. It can also be closely approximated by a linear $\pi/2$ -shifted BPSK modulation with a special pulse shaping filter [7]. Note that at the receiving end, the constant-envelope property of the desired signal can be somewhat distorted due to the effect of channel dispersion and receiver filtering.

In the EDGE air interface, the GMSK modulation can be combined with four different coding rates ranging from 0.37 to 1, resulting in four different modulation/coding schemes, denoted MCS-1 through MCS-4, respectively. Each coded block is interleaved across four data bursts. The burst format is identical with that of GSM. Each burst contains a training sequence of 26 symbols in the middle, 58 data symbols on each side of the training sequence, 3 known tail symbols at either end of the burst, and 8.25 guard symbols at the end, for a total of 156.25 symbols. The symbols are transmitted at 13/48 MHz, resulting in a burst duration of about 0.577 ms.

All simulation results are produced with the DC offset set at 40 dB above the averaged received power of the desired signal (i.e. $d/\sigma_s = 40\text{dB}$). The DC offset is estimated independently from one burst to another. The number of received signal samples used to perform blind DC estimation is $N = 156$. The receiver filter is a root-raised-cosine Nyquist filter with roll-off 0.3.

Figure 1 shows the performance of the proposed DC estimation algorithm with a (static) AWGN channel in terms of the mean-squared estimation error (MSE) given by $E|d - \hat{d}|^2$, where d and $\hat{d}$ denote the actual DC offset and the DC-offset estimate computed over each burst, respectively. As shown in this figure, the proposed algorithm outperforms the simple sample average at all SNR levels. The reduction in estimation error is especially significant in the higher SNR region where

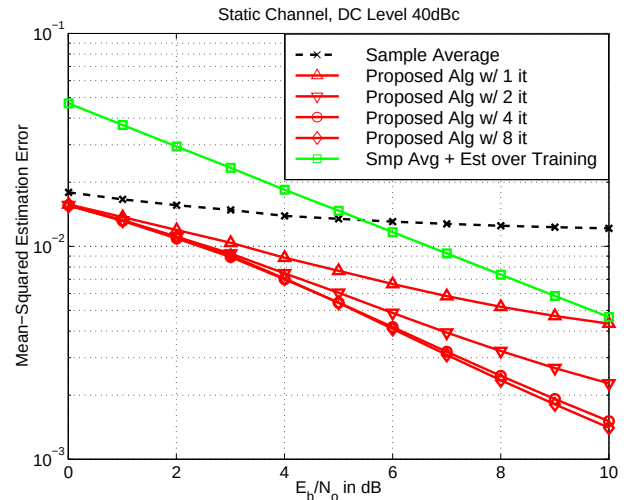


Fig. 1. Mean-Squared Error Performance with AWGN Channel

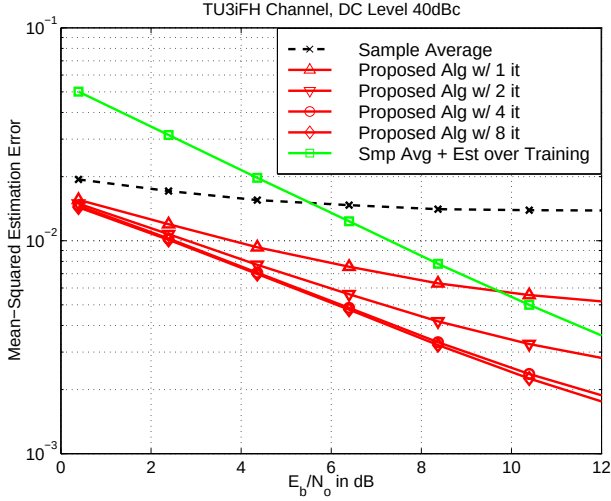


Fig. 2. Mean-Squared Error Performance with TU Fading Channel

the non-Gaussian distributed signal dominates. Only about 4 iterations are needed to achieve most of the gain.

For comparison, we also show the performance of a hybrid scheme, where the training-based DC estimation method described in [1] is combined with the simple sample average. In this scheme, the sample average of the received signal over the entire burst is first computed and subtracted from the received signal. The resulting signal is then used to compute a finer DC estimate over the training sequence using the algorithm in [1]. The effective DC estimate is the sum of the fine DC estimate and the sample average. Since the knowledge of the training sequence allows the receiver to effectively remove the influence of the desired signal from the received signal, the estimation error decreases linearly as SNR increases.

As shown in Figure 1, the additional training-based DC estimation can actually degrade the performance of the sample average at low SNR region, due to the limited length of the training sequence. For a sufficient number of iterations, the proposed blind estimation algorithm consistently outperforms the hybrid scheme at all observed SNR levels.

Figure 2 shows a similar plot for the case with the Typical Urban (TU) channel of GSM at a vehicle speed of 3 km/hr

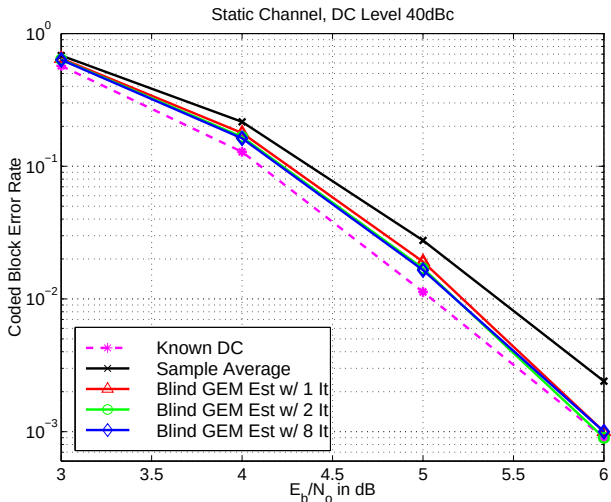


Fig. 3. Coded Block Error Performance with AWGN Channel

with carrier frequency of 900MHz. The gain in MSE over the sample average of received signal is somewhat reduced because the dispersion of the TU channel distorts the constant-envelope property of the transmitted signal, and the signal model adopted in Section VI becomes less accurate. Nevertheless, the proposed algorithm still significantly outperforms the sample average and the hybrid estimation scheme.

Figure 3 compares the coded block error rate performance of receivers that employ different methods of DC compensation. The desired signal is transmitted with the GMSK modulation and MCS-3 coding. The channel is static and non-dispersive. At the receiving end, the DC-offset estimate is first computed and subtracted from the received signal. Conventional LS channel estimation and MLSE equalization are subsequently performed on the resulting DC-compensated signal. As shown in this figure, the proposed algorithm provides a gain of about 0.3 dB in receiver sensitivity at 10% block error rate, which is only about 0.1 dB away from the ideal performance with perfect DC compensation. Moreover, only 2 iterations are needed to achieve such performance.

VIII. CONCLUSION

A method of blindly estimating DC offset based on a block of received signal samples was proposed for use in direct conversion receivers. Unlike the conventional method of taking the sample average of the received signal, which implicitly assumes that the underlying modulated signal is Gaussian distributed, this method exploits the non-Gaussian statistical distribution of the desired modulated signal in order to improve the DC estimate. Two iterative algorithms were derived for jointly estimating the DC offset, the noise variance and the parameters of the non-Gaussian signal distribution. One of these algorithms was applied to the special case where the modulated signal has a constant amplitude. Simulation results generated using GMSK modulated signals showed that the proposed method, which has relatively low implementation complexity, can provide significant performance gain in estimating DC offset in practical receivers.

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